

Real-Time GPA Confidence Scoring

for AI-Generated Results

gpa-scoring module | Snowflake Cortex AI

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Framework:	Agent GPA (Goal-Plan-Action) – arXiv: 2510.08847
Judge model:	claude-haiku-4-5 via SNOWFLAKE.CORTEX.COMPLETE
Scoring:	Goal Fulfillment (GF) + Logical Consistency (LC) – 0-3 each
Storage:	SIGNAL_DB.META_DATA.GPA_SCORE_HISTORY

3 – High	2 – Good	1 – Partial	0 – Low
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Scores appear inline after every chat response – non-blocking, fire-and-forget

Overview

Every AI-generated response in the Cortex Agent Application is automatically evaluated by an LLM judge (claude-haiku-4-5) the moment streaming completes. The judge scores two dimensions - Goal Fulfillment (GF) and Logical Consistency (LC) - and displays colour-coded badges inline alongside each response. Scores are persisted to Snowflake for longitudinal quality tracking.

The system is based on the Agent GPA (Goal-Plan-Action) Framework from Snowflake AI Research (arXiv: 2510.08847) and is implemented as a reusable, agent-agnostic module (gpa-scoring/) that can be dropped into any Next.js + Snowflake application.

End-to-End Architecture

```

Browser (Next.js)
|
+-- 1. User sends question
|   |
|   ?
|   Cortex Agent streams response (SSE)
|   |
|   ?
+-- 2. Stream completes - message rendered
|   |
|   ? (async, non-blocking)
+-- 3. POST /api/agent/score
|   | { question, response, threadId, messageId }
|   |
|   ? (server-side)
|   gpa-scoring module
|   +-- response-prep.ts -> tiered handling (full/extract/summarise)
|   +-- scorer.ts        -> build prompt + call CORTEX.COMPLETE
|   +-- storage.ts       -> persist to GPA_SCORE_HISTORY
|   |
|   ?
+-- 4. Returns GpaScore JSON
|   |
|   ?
+-- 5. GpaScoreBadge renders
    GF: High (3/3)    LC: Good (2/3)
    [warning banner if GF < 2]
  
```

The gpa-scoring Module

The scoring logic is fully extracted into a standalone, reusable module:

```
gpa-scoring/
+-- lib/types.ts           All interfaces, rubrics, config types
+-- lib/scorer.ts          Core scoring function (pure, no framework deps)
+-- lib/response-prep.ts   Tiered response handling
+-- lib/storage.ts         Snowflake persistence
+-- api/score.route.ts     Next.js handler factory
+-- components/
|   +-- GpaScoreBadge.tsx  Badge + tooltip + warning (zero external deps)
+-- hooks/
|   +-- use-gpa-scoring.ts useGpaScoring + useBatchGpaScoring
+-- sql/
    +-- setup.sql          DDL + analytics queries
```

Any agent application integrates by:

- Copying gpa-scoring/ into the project
- Creating one API route file with a config
- Adding the hook and badge to the chat component

Scoring Dimensions

Currently Implemented

Dim	Full Name	Requires Trace	Scale
GF	Goal Fulfillment	No	0-3
LC	Logical Consistency	No	0-3

GF and LC work from the response text alone - available immediately after streaming. Both are evaluated in one LLM call (cost-efficient).

Score Labels

Score	Label	Color	Meaning
3	High	Green	Excellent - fully answered, no contradictions
2	Good	Teal	Mostly correct - minor omissions
1	Partial	Yellow	Incomplete - missing key components
0	Low	Red	Failed - incorrect or refused to answer

Step-by-Step Implementation

Step 1 - Tiered Response Preparation (response-prep.ts)

Long responses are preprocessed before scoring to stay within token budget while preserving the highest-value content:

Response length	-> Strategy
? 8,000 chars	-> Use full response as-is
8,001 – 30,000 chars	-> Structured extraction (no LLM call)
> 30,000 chars	-> LLM summarisation (one Haiku call)

Structured extraction pulls four sections from the response:

- [OPENING] - first 1,500 chars (direct answer usually here)
- [KEY FINDINGS] - sentences containing PV terms: PRR, ROR, MedDRA, NCT IDs, Bradford-Hill, GVP, PRAC, etc.
- [STRUCTURED POINTS] - any numbered list items (conclusions/actions)
- [CONCLUSION] - last 1,500 chars

This approach captures ~90% of evaluation-relevant content for typical clinical safety responses (3,000-15,000 chars) without any extra LLM call.

Step 2 - Prompt Construction (scorer.ts)

A single prompt evaluates all configured dimensions simultaneously:

You are evaluating an AI agent response on the following dimensions.

Question: [truncated to 1,000 chars]

Agent Response [extracted from 12,340 chars]:

[OPENING]

...

[KEY FINDINGS]

...

[CONCLUSION]

...

GF – Goal Fulfillment: Did the agent completely answer the question?

3 = Fully answered with complete, accurate results

2 = Mostly answered with minor omissions

1 = Partially answered, missing key components

0 = Failed, incorrect, or refused to answer

LC – Logical Consistency: Are reasoning steps internally consistent?

3 = All reasoning consistent, no contradictions

2 = Minor inconsistencies that do not affect outcome

1 = Noticeable inconsistencies or contradictory statements

0 = Major logical contradictions

Return ONLY valid JSON:

```
{"gf_score": <0-3>, "gf_reasoning": "<one sentence>",
  "lc_score": <0-3>, "lc_reasoning": "<one sentence>"}
```

Step 3 - Scoring Execution

```
// scorer.ts - simplified view
const result = await SNOWFLAKE.CORTEX.COMPLETE('claude-haiku-4-5', prompt);
const parsed = JSON.parse(result.match(/\{[\s\S]*\}/)[0]);

return {
  GF: { value: clamp(parsed.gf_score), label: toLabel(parsed.gf_score),
        reasoning: parsed.gf_reasoning },
  LC: { value: clamp(parsed.lc_score), label: toLabel(parsed.lc_score),
        reasoning: parsed.lc_reasoning },
};
```

Step 4 - Persistence (storage.ts)

Fire-and-forget: the score is written to Snowflake after the HTTP response is sent (never blocks the user):

```
INSERT INTO SIGNAL_DB.META_DATA.GPA_SCORE_HISTORY (
  agent_name, thread_id, message_id,
  question, response_preview,
  gf_score, gf_label, gf_reasoning,
  lc_score, lc_label, lc_reasoning,
  tier, duration_ms
) VALUES (...)
```

The tier column records which preparation strategy was used (full, extracted, or summarised).

API Route - Handler Factory

The `createScoreHandler()` factory returns a fully configured Next.js POST handler. The `clinical-trial-app` mounts it in 26 lines:

```
// app/api/agent/score/route.ts
import { query } from "@lib/snowflake";
import { createScoreHandler, registerSnowflakeFns }
  from "@gpa-scoring/api/score.route";

registerSnowflakeFns(
  (sql) => query<{ RESULT: string }>(sql), // for scoring
  (sql) => query(sql)                     // for storage
);

export const POST = createScoreHandler({
  agentName: "CLINICALTRIALSAFETYANALYZER",
  judgeModel: "claude-haiku-4-5",
  dimensions: ["GF", "LC"],
  storage: {
    table: "SIGNAL_DB.META_DATA.GPA_SCORE_HISTORY",
    enabled: true,
  },
  lowConfidenceThreshold: 2,
});
```


React Integration

Hook (use-gpa-scoring.ts)

```
// In your chat hook - after streaming completes:
const { score, isScoring, triggerScore } = useGpaScoring({
  scoreEndpoint: "/api/agent/score",
  threshold: 2,
  onScore: (score) => updateMessage(msgId, { gpaScore: score }),
});

// Fire after stream ends:
triggerScore({ question, response, threadId, messageId });
```

Badge Component (GpaScoreBadge.tsx)

```
// In MessageBubble - shows badges below every assistant message:
<GpaScoreBadge
  score={message.gpaScore}
  threshold={2}
  dimensions={["GF", "LC"]}
/>
```

The badge component has zero external dependencies - uses only inline styles. Works in any React project without Tailwind or shadcn.

UX Behaviour

State	Visual
Streaming	No score badge
Scoring in-flight	Grey "Scoring..." spinner
GF >= 2	GF: High (3/3) LC: Good (2/3) - coloured badges with tooltips
GF < 2	Badges + amber warning: "Low confidence - try rephrasing..."
Score API error	Badges disappear silently
Batch re-score	All badges show spinners, update simultaneously

Batch Re-Scoring

A "Re-evaluate All" button rescores every previous message in the thread:

```
const { isRescoring, rescoreAll } = useBatchGpaScoring({
  scoreEndpoint: "/api/agent/score",
});

// On button click:
rescoreAll(messages, threadId, (id, score) => {
  updateMessage(id, { gpaScore: score });
});
```

Uses Promise.allSettled - all messages scored in parallel. Useful after changing agent instructions or semantic model to verify quality improvement.

Configuration Options

Option	Default	Description
agentName	required	Label stored in score history
judgeModel	claude-haiku-4-5	Any CORTEX.COMPLETE model
dimensions	["GF", "LC"]	Add "EE", "PQ", "PA" for full GPA
maxQuestionChars	1000	Question truncation limit
responseTiers.fullUpTo	8000	Use full response up to this length
responseTiers.extractUpTo	30000	Extract up to this length
storage.table	GPA_SCORE_HISTORY	Fully qualified table name
storage.enabled	true	Set false to disable persistence
lowConfidenceThreshold	2	GF below this shows warning banner

Performance Characteristics

Metric	Value
Scoring latency	~2-3 seconds after response
LLM calls per response	1 (both GF and LC in one prompt)
Token budget (typical)	~800-2,000 tokens
Blocks user?	No - fire-and-forget
Blocks response return?	No - persistence is async
Error handling	Silent degradation - badges disappear
Tiered handling	Auto-selected based on response length

Snowflake Storage Schema

```
CREATE TABLE IF NOT EXISTS SIGNAL_DB.META_DATA.GPA_SCORE_HISTORY (
  id          VARCHAR DEFAULT UUID_STRING() PRIMARY KEY,
  agent_name  VARCHAR,      -- which agent was scored
  thread_id   VARCHAR,      -- conversation thread ID
  message_id  VARCHAR,      -- specific message ID
  question    TEXT,         -- user question (max 500 chars)
  response_preview TEXT,    -- response preview (max 500 chars)
  gf_score    NUMBER,
  gf_label    VARCHAR,
  gf_reasoning TEXT,
  lc_score    NUMBER,
  lc_label    VARCHAR,
  lc_reasoning TEXT,
  ee_score    NUMBER,      -- NULL unless EE dimension enabled
  ee_label    VARCHAR,
  ee_reasoning TEXT,
  pq_score    NUMBER,      -- NULL unless PQ dimension enabled
  pq_label    VARCHAR,
  pq_reasoning TEXT,
  pa_score    NUMBER,      -- NULL unless PA dimension enabled
  pa_label    VARCHAR,
  pa_reasoning TEXT,
  tier         VARCHAR,      -- full / extracted / summarised
  duration_ms NUMBER,      -- scoring latency
  scored_at   TIMESTAMP_LTZ DEFAULT CURRENT_TIMESTAMP()
);
```

Analytics Queries

Average scores by agent and day

```
SELECT
    agent_name,
    DATE_TRUNC('day', scored_at)          AS day,
    COUNT(*)                             AS total_scored,
    ROUND(AVG(gf_score), 2)              AS avg_gf,
    ROUND(AVG(lc_score), 2)              AS avg_lc,
    SUM(CASE WHEN gf_score < 2 THEN 1 ELSE 0 END) AS low_confidence
FROM SIGNAL_DB.META_DATA.GPA_SCORE_HISTORY
GROUP BY 1, 2
ORDER BY day DESC, agent_name;
```

Low-scoring responses to investigate

```
SELECT agent_name, thread_id, question,
       gf_score, lc_score, gf_reasoning, lc_reasoning
FROM SIGNAL_DB.META_DATA.GPA_SCORE_HISTORY
WHERE gf_score < 2 OR lc_score < 2
ORDER BY scored_at DESC
LIMIT 20;
```

Thread-level quality summary

```
SELECT agent_name, thread_id,
       COUNT(*)           AS messages,
       ROUND(AVG(gf_score), 2) AS avg_gf,
       ROUND(AVG(lc_score), 2) AS avg_lc
FROM SIGNAL_DB.META_DATA.GPA_SCORE_HISTORY
WHERE thread_id IS NOT NULL AND thread_id != ''
GROUP BY 1, 2
ORDER BY avg_gf ASC
LIMIT 20;
```

Scoring tier distribution (response length analysis)

```
SELECT agent_name, tier,
       COUNT(*)           AS responses,
       ROUND(AVG(duration_ms), 0) AS avg_latency_ms,
       ROUND(AVG(gf_score), 2) AS avg_gf
FROM SIGNAL_DB.META_DATA.GPA_SCORE_HISTORY
GROUP BY 1, 2
ORDER BY agent_name, tier;
```

Integrating with Other Agents

The gpa-scoring module is agent-agnostic. Follow these steps to add real-time scoring to any Cortex Agent in a Next.js application.

Step 1 - Copy the Module

```
cp -r gpa-scoring/ your-new-agent-app/
```

The folder goes at the root of your Next.js app so `@/gpa-scoring/...` resolves via `tsconfig.json` paths.

Step 2 - Create the Snowflake Table (once per account)

Run `gpa-scoring/sql/setup.sql` in Snowsight. All agents share one table, partitioned by `agent_name`:

```
CREATE TABLE IF NOT EXISTS SIGNAL_DB.META_DATA.GPA_SCORE_HISTORY (
  id VARCHAR DEFAULT UUID_STRING() PRIMARY KEY,
  agent_name VARCHAR, thread_id VARCHAR, message_id VARCHAR,
  question TEXT, response_preview TEXT,
  gf_score NUMBER, gf_label VARCHAR, gf_reasoning TEXT,
  lc_score NUMBER, lc_label VARCHAR, lc_reasoning TEXT,
  tier VARCHAR, duration_ms NUMBER,
  scored_at TIMESTAMP_LTZ DEFAULT CURRENT_TIMESTAMP()
);
```

Step 3 - Create the Score API Route

One file per agent. Only `agentName` changes between agents:

```
// app/api/agent/score/route.ts
import { query } from "@lib/snowflake";
import { createScoreHandler, registerSnowflakeFns }
  from "@gpa-scoring/api/score.route";

registerSnowflakeFns(
  (sql) => query<{ RESULT: string }>(sql),
  (sql) => query(sql)
);

export const POST = createScoreHandler({
  agentName: "YOUR_AGENT_NAME", // only change this per agent
  judgeModel: "claude-haiku-4-5",
  dimensions: ["GF", "LC"],
  storage: {
    table: "SIGNAL_DB.META_DATA.GPA_SCORE_HISTORY",
    enabled: true,
  },
  lowConfidenceThreshold: 2,
});
```

Step 4 - Add the Hook to Your Chat Logic

Fire scoring after each response finishes streaming:

```
// hooks/use-chat.ts
import { useGpaScoring } from "@gpa-scoring/hooks/use-gpa-scoring";

const { triggerScore } = useGpaScoring({
  scoreEndpoint: "/api/agent/score",
  threshold: 2,
  onScore: (score) => {
    setMessages(prev => prev.map(m =>
      m.id === currentMsgId ? { ...m, gpaScore: score } : m
    ));
  },
});

// Call when streaming completes:
triggerScore({ question, response, threadId, messageId });
```

Step 5 - Render the Badge

Add below each assistant message:

```
// components/MessageBubble.tsx
import { GpaScoreBadge } from "@gpa-scoring/components/GpaScoreBadge";

{message.role === "assistant" && (
  <GpaScoreBadge score={message.gpaScore} threshold={2} />
)}
```

Step 6 - Optional: Batch Re-Score Button

```
import { useBatchGpaScoring } from "@gpa-scoring/hooks/use-gpa-scoring";

const { isRescoring, rescoreAll } = useBatchGpaScoring({
  scoreEndpoint: "/api/agent/score",
});

// On button click:
rescoreAll(messages, threadId, (id, score) => {
  setMessages(prev => prev.map(m =>
    m.id === id ? { ...m, gpaScore: score } : m
  ));
});
```

Files Changed Per Agent

File	What to change	What stays the same
app/api/agent/score/route.ts	agentName	All other config
Your chat hook	scoreEndpoint URL if different	triggerScore call pattern
Your message component	Nothing	GpaScoreBadge unchanged
gpa-scoring/ folder	Nothing	Shared module for all agents

Integration Checklist

- Copy gpa-scoring/ into the app
- Run sql/setup.sql in Snowsight (once per account)
- Create app/api/agent/score/route.ts with your agentName
- Call triggerScore() after streaming completes
- Add GpaScoreBadge to your message component
- Test: send a message, badge appears 2-3 seconds later
- Verify: `SELECT * FROM GPA_SCORE_HISTORY WHERE agent_name = 'YOUR_AGENT'`

References

- Agent GPA Paper: "What is Your Agent's GPA? A Framework for Evaluating Agent Goal-Plan-Action Alignment" - arXiv: 2510.08847
- Snowflake Cortex COMPLETE: <https://docs.snowflake.com/en/sql-reference/functions/complete-snowflake-cortex>
- Module location: gpa-scoring/ - reusable across any Next.js + Snowflake project